

CHAPTER 5

EMOTIONAL INTERACTION

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Objectives

The main aims of this chapter are to:

- Explain how our emotions relate to behavior and user experience.
- Provide examples of interfaces that are both pleasurable and usable.
- Explain what expressive and annoying interfaces are and the effects they can have on people.
- Introduce the area of automatic emotion recognition and emotional technologies.
- Describe how technologies can be designed to change people's attitudes and behavior.
- Give an overview on how anthropomorphism has been applied in interaction design.
- Enable you to critique the persuasive impact of an online agent on customers.





5.1 Introduction

An overarching goal of interaction design is to develop products that elicit positive responses from users, such as feeling at ease, being comfortable, and enjoying the experience of using them – be it a washing machine or a flight deck. Designers are also concerned with how to create interactive products that elicit specific kinds of emotional responses in users, such as motivating them to learn, play, or be creative or social. There has also been much interest in designing websites and apps that people can trust, and that make them feel comfortable about divulging personal information when making a purchase or giving feedback.

Taken together, we refer to this emerging area as emotional interaction. In this chapter we look at how and why the design of interactive products may cause certain kinds of emotional responses in people. We begin by explaining what emotions are and how they shape our behavior and everyday experiences. We then look at expressive interfaces, examining the role of an interface's appearance to users and how it affects usability and the user experience. We then consider how products elicit positive effects, e.g. pleasure, or negative responses, e.g. frustration. We introduce technological approaches to sensing people's emotions and how these are being used to inform the design of new kinds of emotional technology. The ways technologies are being designed and used to

persuade people to change their behavior and attitudes are then covered. We look, in particular, at ubiquitous technology interventions that are being designed to improve health and well-being and reduce domestic energy and water consumption. Following this, we show how anthropomorphism has been used in interaction design and the implications of designing applications that have human-like qualities. To illustrate this approach, virtual characters and robot pets are described that have been developed to motivate people to learn, buy, and listen.

5.2 Emotions and the User Experience

Emotional interaction is concerned with how we feel and react when interacting with technology. It covers different aspects of the user experience, from how we feel when first finding out about a new product to getting rid of it. It also looks at why people become emotionally attached to certain products (e.g. virtual pets), how social robots might help reduce loneliness, and how to change human behavior through the use of emotive feedback.

Consider the different emotions you experience for a common everyday activity – shopping online for a product, such as a new smartphone, a washing machine, or a vacation. Firstly, there is the realization of needing or wanting it, and then the desire and anticipation of purchasing it. This is followed by the joy or frustration of finding out more about what products are available and deciding which to choose from potentially hundreds or even thousands (by visiting numerous websites, such as comparison sites, reviews, recommendations, and social media sites). This entails matching what is available with what you like or need and whether you can afford it. The thrill of deciding on a purchase may be quickly followed by the shock of how much it costs and the disappointment that you can't afford it. The process of having to decide again may be accompanied by annoyance as you can't find one that is as good as your first choice. You think about other options, such as seeking advice from an expert in a shopping mall, but you have an aversion to sales assistants and don't trust their advice, because you think

they have their own interests (making money), rather than yours, at heart. So you carry on looking, getting more tired and frustrated. When you do make a decision, you experience a sense of relief. You click merrily through the various options (such as color, size, warranty) and then the dreaded online payment form pops up. You type in all your details and press the final payment button. A window then appears saying that your credit card number is incorrect. So you type it in again very slowly. And you notice you need to type the three-number security code in again. Finally, when all is done you let out a big sigh. But as you walk away from your computer, doubts start to form in your mind – maybe you should have bought the other one ...

This rollercoaster set of emotions is what many of us experience when shopping online, especially for expensive products, where there is a myriad of options to choose from and where we want to be sure that we make the right choice.

Activity 5.1

Wufoo is a company specializing in building online forms, with the intention of transforming what are usually boring and tedious tasks into more fun activities. How do the forms in [Figure 5.1](#) compare with others you have had to fill in?

The image displays two side-by-side examples of Wufoo online forms. Both forms have a red header with the 'Wufoo' logo.

Left Form: Customer Satisfaction Survey

- Title: Customer Satisfaction Survey
- Introductory text: Please take a few moments to complete this satisfaction survey.
- Section 1: Overall, how satisfied were you with the product / service?
 - Options: ☐ Very Satisfied, ☐ Satisfied, ☐ Neutral, ☐ Unsatisfied, ☐ Very Unsatisfied, ☐ N/A
- Section 2: Would you recommend our product / service to colleagues or contacts within your industry?
 - Options: ☐ Definitely, ☐ Probably, ☐ Not Sure, ☐ Probably Not, ☐ Definitely Not
- Section 3: Would you use our product / service in the future?
 - Options: ☐ Definitely, ☐ Probably, ☐ Not Sure, ☐ Probably Not, ☐ Definitely Not
- Section 4: How long have you used our product / service?
 - Options: ☐ Less than a month, ☐ 1-6 months, ☐ 1-3 years, ☐ Over 3 Years, ☐ Never used

Right Form: Workshop Registration

- Title: Workshop Registration
- Introductory text: Register now while seats are available!
- Section 1: Name *
 - Fields: Title, First, Last, Suffix
- Section 2: Address *
 - Fields: Street Address, Address Line 2
- Section 3: City, State / Province / Region, Postal / Zip Code, Country
- Section 4: Email *
- Section 5: Lunch
 - Text: At noon, we will break for a one hour lunch.
 - Text: Please select a meal preference.
 - Field: Lemon Chicken (dropdown menu)
- Section 6: Phone Number
 - Fields: (3 digits) - (3 digits) - (4 digits)
- Submit button

Figure 5.1 Examples of Wufoo's online forms

Source: Reproduced with permission from [Wufoo.com](http://wufoo.com/examples/#survey).
<http://wufoo.com/examples/#survey>.

Comment

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Emotional interaction is about considering what makes us happy, sad, annoyed, anxious, frustrated, motivated, delirious, and so on, and using this knowledge to inform the design of different aspects of

the user experience, from when we first want something to when we no longer interact with it or need to replace it. However, it is not straightforward to achieve as people's moods and feelings are constantly changing. There are also many reasons that might cause someone to be happy or sad, such as the sun shining or someone else winning a game.

A good place to start understanding how emotions affect behavior and how behavior affects emotions is to examine how people express themselves and read each other's expressions. These include understanding the relationship between facial expressions, body language, gestures, and tone of voice. For example, when people are happy they typically smile, laugh, and open their bodies up. When they are angry they shout, gesticulate, and screw up their face. A person's expressions can trigger emotional responses in others. So when someone smiles it can cause others to feel good and smile back. Emotional skills, especially the ability to express and recognize emotions, are central to human communication. Most of us are highly skilled at detecting when someone is angry, happy, sad, or bored by recognizing their facial expressions, way of speaking, and other body signals. We are also very good at knowing what emotions to express in a given situation. For example, when someone has just heard he has failed an exam, we know it is not a good time to smile and be happy. Instead we try to empathize.

So what do you do when you are in a bad mood? Or when you suddenly get scared? There is an ongoing debate about how and whether emotion causes behavior. For example, does being angry make you concentrate better? Or does being happy make you take more risks, such as spending too much money? Or vice versa? Or neither? It could be that we can just be happy, sad, or angry and that this does not affect our behavior. Many theorists argue that emotions cause behavior, for example the 'fear brings flight and anger brings fight' perspective. When we are frightened or angry, the emotional response is to focus on the problem at hand and try to overcome or resolve the perceived danger. Our bodies will respond by tensing our muscles and sweating. In contrast, when we are very happy, such as watching our team win the last game of the championship, the

emotional response is to laugh, cheer, and jump about. The body relaxes.

Baumeister *et al* (2007) argue that the role of emotion is more complicated than a simple cause and effect model; emotions can be both simple and short-lived or complex and long-lasting. For example, someone can become startled by a sudden, unexpected loud noise or remain annoyed for hours when staying in a hotel room that has a noisy air conditioning unit. To distinguish between these types of emotions and their effect on behavior, automatic responses (e.g. being startled) are called affect. They can happen rapidly, typically within a fraction of a second and, likewise, may dissipate just as quickly. Conscious emotions, on the other hand, tend to be slow to develop and equally slow to dissipate, and are often the result of a conscious cognitive behavior, such as weighing up the odds, reflecting, or contemplating. They require our attention and underlie our ability to learn and adapt our behavior. For example, someone might drink too much when out at a bar one night with a friend, causing them to become rowdy, which the next morning makes them feel guilty and embarrassed. These subsequent feelings make them reflect on their behavior, such that they realize they have behaved inappropriately. This could then lead them to think about how to deal with such a situation in the future, which might be to have only non-alcoholic drinks when going to the pub with this friend.

Understanding how emotions work provides a way of considering how to design interfaces and apps that can trigger affect or reflection in the user. For example, Norman (2005) suggests that being in a positive state of mind can enable us to be more creative as we are less focused. Designers, therefore, might consider how to design products that can make people feel happy, assuming it will then make them creative. He also suggests that when people are happy, they are more likely to overlook and cope with minor problems they are experiencing with a device. In contrast, when someone is anxious or angry, they are more likely to be less tolerant. So if the product is intended to be used during leisure time and is meant to be fun and enjoyable to use, designers “can get away with more” and not be too worried about how the information appears at the

interface. On the other hand, he says that for serious tasks, such as monitoring a process control plant or driving a car, designers need to pay special attention to all the information required to do the task at hand and that the interface should be visible with clear and unambiguous feedback. The bottom line is “things intended to be used under stressful situations require a lot more care, with much more attention to detail” (Norman, 2005, p. 26).

Activity 5.2

Do you feel more creative when you are in a happy mood? Do you get less work done when you are feeling stressed?

Comment

Show/Hide

Video of Kia Höök talking about affective interaction in a series of videos that consider emotion in terms of how it is constructed in interaction with technology and people is available at

www.interaction-design.org/encyclopedia/affective_computing.html

BOX 5.1

A Model of Emotional Design

Ortony *et al*'s (2005) model of emotion and behavior is couched in terms of different levels of the brain. At the lowest level are parts of the brain that are pre-wired to automatically respond to events happening in the physical world. This is called the visceral level. At the next level are the brain processes that control our everyday behavior. This is called the behavioral level. At the highest level are brain processes that contemplate. This is called the reflective level ([Figure 5.2](#)). The visceral level responds rapidly, making judgments about what is good or bad, safe or dangerous, pleasurable or abhorrent. It also triggers the emotional responses to stimuli (e.g. fear, joy, anger, and sadness) that are expressed through a combination of physiological and behavioral responses. For example, on seeing a very large hairy spider running across the floor of the bathroom, many people will experience fear, causing them to scream and run away. The behavioral level is the site where most human activities occur; examples include well-learned routine operations such as talking, typing, and driving. The reflective level entails conscious thought where people generalize across events or step back from the routine and the immediate. An example is switching between thinking about the narrative structure and special effects used in a Harry Potter movie and becoming scared at the visceral level when watching the movie. ■

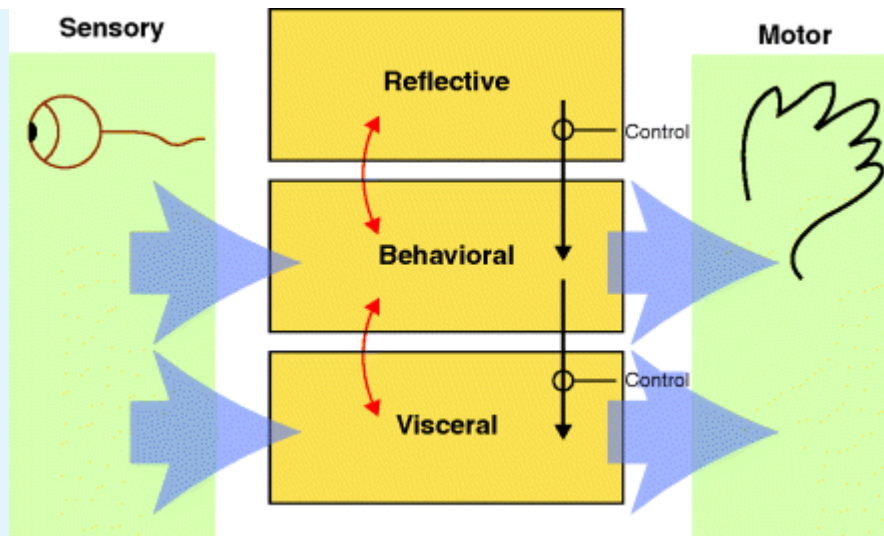


Figure 5.2 Ortony et al's (2005) model of emotional design showing three levels: visceral, behavioral, and reflective

Source: The illustration and text are from [Figure 1.1](#) of Norman, D. A. (2004). *Emotional Design: We love (or hate) everyday things*. New York: Basic Books. Reprinted with permission of the author.

One way of using the model is to think about how to design products in terms of the three levels. Visceral design refers to making products look, feel, and sound good. Behavioral design is about use and equates with the traditional values of usability. Reflective design is about taking into account the meaning and personal value of a product in a particular culture. For example, the design of a Swatch watch focuses on reflective aspects, where the aesthetics and the use of cultural images and graphical elements are central. Brilliant colors, wild designs, and art are very much part of the Swatch trademark and are what draw people to buy and wear their watches.



Figure 5.3 A Swatch watch called Dip in Color

Source: With permission from The Swatch Group (UK) Limited.
<http://store.swatch.com/suop103-dip-in-color.html>



“it’s a very user-friendly model.”

5.3 Expressive Interfaces

Expressive forms like emoticons, sounds, icons, and virtual agents have been used at the interface to (i) convey emotional states and/or (ii) elicit certain kinds of emotional responses in users, such as feeling at ease, comfort, and happiness. Icons and animations have been used to indicate the current state of a computer or a phone, notably when it is waking up or being rebooted. A classic from the 1980s and 1990s was the happy Mac icon that appeared on the screen of the Apple computer whenever the machine was booted (see [Figure 5.4](#)). The smiling icon conveyed a sense of friendliness, inviting the user to feel at ease and even smile back. The appearance of the icon on the screen was also very reassuring to users, indicating that their computer was working correctly. This was especially true for situations where users had to reboot their computer after it had crashed, and where previous attempts to reboot had failed (usually indicated by a sad icon face – see [Figure](#)

[5.4](#)). After nearly 20 years, the happy Mac icon was laid to rest. The sad Mac icon lasted a bit longer, showing its face on an iPod when its software needed restoring. Apple has since switched to the use of more abstract icons to indicate starting up and busy with a process, showing a swirling clock or a colorful beach ball. Android has a lime-green robot icon on its start-up page, suggesting something between being techno-like and human-like.

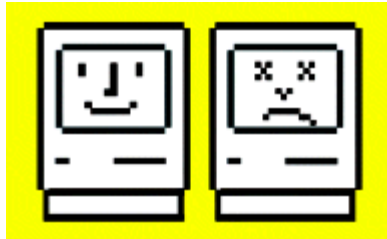


Figure 5.4 Smiling and sad Apple icons for the classic Mac

Other ways of conveying the status of a system are through the use of:

- Dynamic icons (e.g. a recycle bin expanding when a file is placed in it and paper disappearing in a puff when emptied).
- Animations (e.g. a beach ball whirling to say the computer is busy).
- Spoken messages, using various kinds of voices, telling the user what needs to be done (e.g. GPS navigation system instructing you politely where to go after having taken a wrong turn).
- Various sonifications indicating actions and events (e.g. whoosh for window closing, schlook for a file being dragged, ding for new email arriving).
- Vibrotactile feedback, such as distinct smartphone buzzes that specifically represent special messages from friends and family.

The style of an interface, in terms of the shapes, fonts, colors, balance, white space, and graphical elements that are used and the way they are combined, can also influence its emotional impact. Use of imagery at the interface can result in more engaging and enjoyable experiences (Mullet and Sano, 1995). A designer can use

a number of aesthetic techniques such as clean lines, balance, simplicity, and texture. The iPod, featured in [Chapter 1](#), exemplifies this approach.

The design of aesthetically pleasing interfaces has become of central concern to interaction design. Empirical studies have shown that the aesthetics of an interface can have a positive effect on people's perception of the system's usability (e.g. Tractinsky, 1997). When the look and feel of an interface is pleasing and pleasurable – e.g. beautiful graphics, nice feel to the way the elements have been put together, well-designed fonts, elegant use of images and color, a good sense of balance – users are likely to be more tolerant, e.g. they may be prepared to wait a few more seconds for a website to download. Furthermore, good-looking interfaces are often more satisfying and pleasurable to use. A key concern, therefore, is to strike a balance between designing aesthetic versus usable interfaces (Tractinsky *et al*, 2000) and pleasurable versus useful ones (Jordan, 2000).

BOX 5.2

A New Breed of Home Technology: The NEST and Sproutling Interfaces

There is an assortment of technologies on the market now that can track, aggregate, and visualize our every behavior – from toothbrushes that check whether you and your family are brushing your teeth correctly to devices that monitor how well a baby is sleeping and what it is feeling when awake. Two that stand out in terms of their appearance are the Nest (nest.com) and the Sproutling (sproutling.com). The designs are simple, round, and use bright colors – this makes them cute and aesthetically pleasing to the eye. The Nest thermostat provides an intelligent way of controlling how your house is heated or cooled. The Sproutling is a band that is wrapped around a baby's ankle that senses heart rate, skin temperature, motion, and position. It communicates with a smartphone app to let parents know if their baby is sleeping soundly or if something is wrong – using cute baby emoticons. These are examples of products designed for families to use in the home and via their smartphones. Both were designed by ex-Apple employees. Do you think their focus on the design aesthetic enhances the user experience and usability? ■



Figure 5.5 (a) The Nest, (b) the Sproutling ankle band (c) the Sproutling smartphone app

Source: With permission from NestLabs.com and sproutling.com.

5.4 Annoying Interfaces

In many situations, computer interfaces may inadvertently elicit negative emotional responses such as anger and disgust. This typically happens when something that should be simple to use or set turns out to be complex. The most common examples are remote controls, printers, digital alarm clocks, and digital TV systems. Getting a printer to work with a new digital camera, trying to switch from watching a DVD to the TV, and changing the time on a digital alarm clock in a hotel can be very trying. Also, fiddly actions that have to be done every day, such as attaching the ends of USB cables between smartphones, laptops, and fit bands, can be irksome, especially if it is not easy to see which way up to insert them.

This does not mean that developers are unaware of such usability problems. Several methods have been devised to help the novice user get set up and become familiarized with a technology. However, these have sometimes backfired, since the design solution itself has ironically become a source of annoyance and frustration. For

example, one technique that was popularized in the 1990s was the use of friendly agents at the interface. The assumption was that novices would feel more at ease with a companion and would be encouraged to try things out after listening, watching, following, and interacting with it. Microsoft pioneered a class of agent-based software, Bob, aimed at new computer users (many of whom were viewed as computer-phobic). The agents were presented as friendly characters, including a pet dog and a cute bunny. An underlying assumption was that having these kinds of agents on the screen would make users feel more comfortable with using the software. An interface metaphor of a warm, cozy living room, replete with fire and furniture, was also provided (see [Figure 5.6](#)) – again intended to convey a comfortable feeling. However, Bob never became a commercial product. Why do you think not?



Figure 5.6 'At home with Bob' software developed for Windows 95. Although now defunct, it has been resurrected affectionately to run on a Virtual PC platform

Source: Microsoft product screenshot reproduced with permission from Microsoft Corporation.

Contrary to the designers' expectations, many people did not like the idea of Bob at all, finding the interface too cute and childish. However, Microsoft did not give up on the idea of making their

interfaces more friendly and developed other kinds of agents, including the infamous Clippy (a paper clip that has human-like qualities), as part of their Windows 98 operating environment. Clippy typically appeared at the bottom of a user's screen whenever the system thought the user needed help carrying out a particular task (see [Figure 5.7](#)). It, too, was depicted as a cartoon character, with a warm personality. This time, Clippy was released as a commercial product but it was not a success. Many Microsoft users found it very trying and intrusive, distracting them from their work. When it was finally retired, numerous websites posted jokes and witty comments, celebrating its demise.

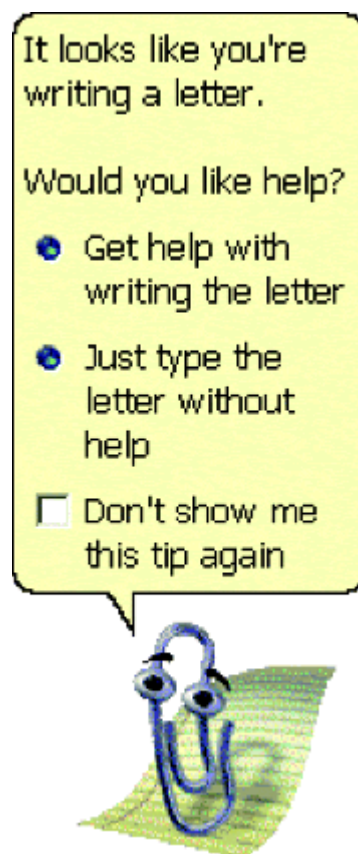


Figure 5.7 Microsoft's agent Clippy

Source: Microsoft product screenshot reproduced with permission from Microsoft Corporation.

Interfaces, if designed poorly, can make people look stupid, or feel insulted or threatened. The effect can be to make them annoyed to

the point of losing their temper. There are many situations that cause such emotional responses. These include:

- When an application doesn't work properly or crashes.
- When a system doesn't do what the user wants it to do.
- When a user's expectations are not met.
- When a system does not provide sufficient information to let the user know what to do.
- When error messages pop up that are vague or obtuse.
- When the appearance of an interface is too noisy, garish, gimmicky, or patronizing.
- When a system requires users to carry out too many steps to perform a task, only to discover a mistake was made somewhere along the line and they need to start all over again.
- Websites that are overloaded with text and graphics, making it difficult to find the information desired and making them slow to access.
- Flashing animations, especially flashing banner ads and pop-up ads that cover what the user is looking at and which require them to actively click on a check box to close them.
- The over-use of sound effects and music, especially when selecting options, carrying out actions, running tutorials, or watching website demos.
- Featuritis – an excessive number of operations, such as the array of buttons on remote controls.
- Poorly laid out keyboards, pads, control panels, and other input devices that cause users to persistently press the wrong keys or buttons.

Activity 5.3

Most of us are familiar with the ‘404 error’ message that pops up now and again when our computer doesn’t upload the web page we’re trying to view. But what does it mean and why the number 404? How does it make you feel when you see it and what do you do next? Is there a better way of letting users know when they or the computer have made an error?

Comment

Show/Hide

Some systems are very emotive in the way they let you know you have made an error. [Figure 5.8](#) shows one such example, where bold letters (that used to be in red) and an exclamation mark are used to alert the user to the severity of the error. The effect can cause a user to become quite anxious, making them panic, especially if subsequently given only two chances to rectify the situation, as is often the case after typing in a password incorrectly. Is it really necessary to be so shouty? Would it not be more pleasant if the message suggested that the user try again? Some companies have started using more polite and useful error messages, such as this one provided for a user who could not log in to his Wallet using the Android app, “Greetings. We’re having a problem. We can’t show the info you requested. Sorry about this. It appears we’re having an unexpected problem getting you what you need. Try again.”

Error!

Your login information is incorrect...

Please verify that you typed in your Last Name, NSF ID, and Password correctly. If you still cannot login, please contact the Administrator regarding your access rights.

[Return To Previous Page](#)

Figure 5.8 An error message that appears if a user types in his or her personal details for accessing the protected website incorrectly

Dilemma

Should Computers Say They're Sorry?

A provocative idea is that computers should apologize when they make a mistake. Reeves and Nass (1996), for example, argue that they should be polite and courteous in the same way that people are to one another. While apologizing is normal social etiquette in human behavior, especially when someone makes a mistake, would you agree that computers should be made to behave in the same way? Would users be as forgiving of computers as they are of one another? For example, what would most users think if, after their computer had crashed, it came up with a spoken or written apology such as, "I am really sorry I crashed. I'll try not to do it again"? Would they think that it was being sincere? Would the apology make them forgive the computer in the way they forgive other people, after receiving such an apology? Or would it have no effect at all? Worse still, would users perceive such messages as vacuous statements and regard them simply as condescending, thereby increasing their level of frustration? It seems people would rather a computer said sorry. An empirical study comparing error messages that apologized versus those that did not were perceived by users to be less frustrating (Park *et al*, 2012). ■

5.5 Detecting Emotions and Emotional Technology

The approach called affective computing develops computer-based systems that try to recognize and express emotions in the same way humans do (Picard, 1998). A long-standing area of research in artificial intelligence and artificial life has been the creation of intelligent robots that are designed to behave like humans and other creatures. An early classic was COG, where a group of researchers attempted to build an artificial two-year-old. An offspring of COG was Kismet (Breazeal, 1999), which was designed to engage in meaningful social interactions with humans.

More recently, an area of research that has become of interest to interaction design is automatic emotion analysis. A number of sensing technologies are now used to automatically measure and analyze users' emotions and, from the data collected, predict aspects of their behavior – for example, what is someone most likely to buy online when feeling sad, bored, or happy. The main techniques and technologies used are: cameras for measuring facial expressions; bio-sensors placed on fingers or palms of the hands to measure galvanic skin response (which is used to infer how anxious or nervous someone is as indicated by an increase in their sweat); and body movement and gestures as detected by motion capture systems or accelerometer sensors placed on various parts of the body. Emotion assessing technology, such as automated facial coding, is gaining popularity in commercial settings, especially in marketing and e-commerce. For example, Affdex[®] emotion analytics and insights software from Affectiva[®] employs advanced computer vision and machine-learning algorithms to catalog a user's emotion reactions to digital content, as captured through a standard webcam, to analyze how engaged the user is with movies, online shopping sites, and ads. Six fundamental emotions are classified based on the face expressions Affdex collects (see [Figure 5.9](#)). These are sadness, happiness, disgust, fear, surprise, and anger. If a user screws up their face when an ad pops up, that suggests they feel disgust, whereas if they start smiling, it suggests they are feeling

happy. The website can then adapt its ad, movie storyline, or content to what it perceives the person needs at that point in their emotional state.

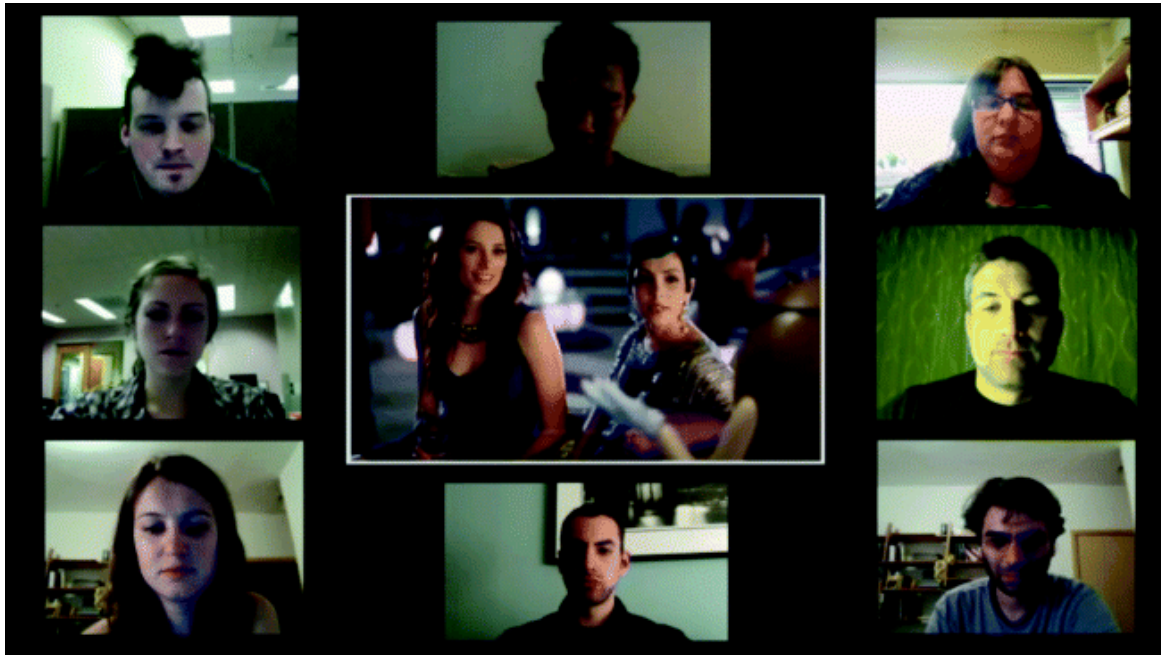


Figure 5.9 A screen shot showing facial coding from Affectiva software

Source: Courtesy of Affectiva, Inc.

Other indirect methods that are used to reveal the emotional state of someone include eye-tracking, finger pulse, speech, and the words/phrases they use when tweeting, chatting online, or posting to Facebook (van den Broek, 2013). The level of affect expressed by users, the language they use, and the frequency with which they express themselves when using social media can all indicate their mental state, well-being, and aspects of their personality (e.g. whether they are extrovert or an introvert, neurotic or calm). Some companies may try to use a combination of these measures, such as facial expression and the language they use when online, while others may focus on just one, such as the tone of their voice, when answering questions over the phone. This kind of indirect emotion detection is beginning to be used to help infer or predict someone's behavior; for example, determining their suitability for a job, or how they will vote at an election.

Dilemma

Is It OK for Technology to Work Out How You Are Feeling?

Do you think it is creepy that technology is trying to read your emotions from your facial expressions or from what you write in your tweets and, based on its analysis, filter the online content you are browsing, such as ads, the news, or a movie, to match your mood?

Human beings will suggest things to each other, often based on what they think the other is feeling. For example, they might suggest a walk in the park to cheer them up. They might also suggest a book to read or a movie to watch. However, many people don't like the idea that a technology might do the same but at a finer level of granularity; for example, suggesting what you might like to eat, watch, or buy, based on how it analyzes your feelings. ■

An example of a more benign emotional technology is 'Moon Phrases' – an app that has been developed to help people reflect upon their emotional well-being (de Choudhury *et al*, 2013). The app allows people to think about their emotional states and feelings via analyzing what they say in their postings on Twitter. The aim is to help them cope better with stress and anxiety by recognizing the triggers that cause them to occur. The tool works by analyzing the way users express themselves in social media – through the types of words, mood hashtags, emoticons, and expressions used and their frequency. It then visualizes this data in terms of a series of moon icons conveying positive and negative affect, representing each day for a period of several months; full moons indicate positivity, while half or quarter moons reflect more negativity. By looking back at their history of moons, users can start to identify patterns that could help

them understand more about themselves and what might be causing their mood swings.

BOX 5.3

How Much Do You Touch?

As well as expressing our emotions through facial, vocal, gesture, and body movements, we also use touch. For example, when someone touches our shoulder or strokes our arm during a conversation, they are often conveying affection towards us. Conversely, when someone shakes or squeezes our arms, they are more likely to be expressing anger or fear. The type of touch, how much pressure is applied, and where on the body it happens can provide clues as to the emotion being experienced by the person who is touching. HCI researchers are investigating how to develop 'touch profiles' based on measuring how users touch screens and consoles when playing video games (e.g. Gao *et al*, 2012). However, there is not a straightforward mapping between one kind of touch and one type of emotion (they can overlap), and current research tries to model and infer the range of someone's emotional experiences over time. These profiles can then provide information about how they feel without interrupting their user experience – which is very valuable feedback for game designers when thinking of how to design engrossing and gripping games. ■

5.6 Persuasive Technologies and Behavioral Change

A diversity of technologies is increasingly being used to draw people's attention to certain kinds of information in an attempt to change what they do or think. Pop-up ads, warning messages, reminders, prompts, personalized messages, and recommendations are some of the methods that are being deployed on computer screens. Fogg (2003) has labeled this phenomenon persuasive

technology; interactive computing systems are deliberately designed to change people's attitudes and behaviors. Traditionally, media such as magazines, newspapers, pamphlets, radio, and TV have been used to persuade people to join a good cause, give up a bad habit, donate money, or buy a product. For example, a picture of a starving child with big round eyes staring out at the reader on the front of a newspaper is commonly used by charities. The effect is to pull at the readers' heartstrings, inducing feelings of guilt and, in so doing, spur them on to sending money.

More recently, interactive techniques have been used to entice, cajole, and persuade people to do something they might not have otherwise done. Successful examples include Amazon's one-click mechanism that makes it so easy and tempting to buy something at their online store, and recommender systems that suggest specific books, hotels, restaurants, etc a reader might want to try based on their previous purchases, choices, and taste. Splash pages to online shopping sites and color images of gorgeous-looking beach and mountain scenes on travel sites are designed to lure people into making impulse purchases.

In addition to using interactive technologies as a more targeted and personalized form of advertising, they can be used to change people's behaviors in non-commercial domains, such as safety, preventative healthcare, fitness, personal relationships, energy consumption, and learning. Here, the emphasis is on changing habits or doing something that will improve an individual's well-being through monitoring their behavior. An early example was Nintendo's Pokémon Pikachu with pedometer attached that was designed to motivate children into being more physically active on a consistent basis. The owner of the digital pet that lives in the device was required to walk, run, or jump each day to keep it alive. The wearer received credits with each step taken – the currency being watts that could be used to buy Pikachu presents. Twenty steps on the pedometer rewarded the player with one watt. If the owner did not exercise for a week, the virtual pet became angry and refused to play anymore. This use of positive rewarding and sulking can be a powerful means of persuasion, given that children often become

emotionally attached to their virtual pets, especially when they start to care for them.

Similarly, the WaterBot system was developed using a special monitoring and feedback device, but for adults as a way of reducing their usage of water in their homes (Arroyo *et al*, 2005). There is much evidence to suggest that people are wasteful with water, often leaving the tap running continuously for long periods of time while cleaning their teeth or washing. The research team thought that the use of monitoring technology could help persuade householders to change their behavior to be more conservative in their water usage. To this end, they used the theory of positive reinforcement to inform their design, which states that activities are likely to be repeated if some kind of reward is given occasionally and randomly (similar to the reward system used in slot machines). A sensor-based system was developed where positive auditory messages and chimes were sounded when the tap was turned off. The water was also lit with a random pattern of color as a reward for consistent water-saving behavior. Two illuminated bar graphs were also presented alongside the tap, showing how much water a person had used relative to others in the household. Here, the idea was to encourage peer pressure and for the members of the household to talk to each other about their water usage. Informal feedback of the prototype system in a small number of people's homes suggested that the most effective method of persuasion was the constantly changing bar graph. It drew people's attention to the tap, leading them to make quick comparisons between their and the others' water consumption. The rewards of chimes and colored water had less impact, especially as their novelty wore off.

Activity 5.4

Watch the two videos:

- i. The Piano Staircase <http://youtu.be/2lXh2n0aPyw>
- ii. The Outdoor Bin <http://youtu.be/cbEKAwCoCKw>

Do you think such playful methods are effective at changing people's behavior?

Comment

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HAPIfork is a device intended to help someone monitor and track their eating habits (see [Figure 5.10](#)). If it detects they are eating too quickly, it will vibrate (similar to the way a smartphone does when on silent mode) and an ambient light will appear at the end of the fork, providing the eater with real-time feedback intended to slow them down. The assumption is that eating too fast results in poor digestion and poor weight control, and that making people aware that they are gobbling their food down can help them think about how to eat more slowly at a conscious level. Other data is collected about how long it took them to finish their meal, the amount of fork servings per minute, and the time between them. These are turned into a dashboard of graphs and statistics so the user can see each week whether their fork behavior is improving.



Figure 5.10 A friend using the HAPIfork in a restaurant when eating a cake

Activity 5.5

What do you think might be problematic when using this kind of dieting device in a restaurant?

Comment

Show/Hide

Nowadays, there are a number of mobile apps and personal tracking devices on the market that are designed to help people monitor various behaviors and then change their behavior based on the data collected and represented back to them. These include fitness and sleep bands, such as FitBit and Jawbone Up, and weight trackers such as smart scales. Similar to HAPIfork, these devices and apps are designed to encourage people to change their behavior by providing them with dashboards of statistics and graphs that show how much exercise, sleep, or food/drink they have had over a day, week, or longer period, compared with what they have done in the

previous day/week/month. This can also be compared, through online leaderboards and charts, with how well they have done in relation to their peers and friends. A survey of how people use such devices in their everyday lives revealed that often people bought them to simply try them out – rather than specifically to change a particular behavior (Rooksby *et al*, 2014). How, what, and when they tracked depended on their interests and lifestyles; some used them as a way of showing how fast they could run during a marathon or cycle on a track, or how they could change their lifestyle to sleep or eat better.

Video of BBC News item showing smart scales being demonstrated at

<http://www.bbc.co.uk/news/technology-24184634>

An alternative approach to automatically collecting quantified data about a behavior is to ask people to write down manually how they are feeling now or to rate their mood, and to then reflect upon how they felt about themselves in the past. A mobile app called Echo, for example, asks people to write a subject line, rate their happiness at that moment, and add a description, photos, and/or videos if they want to (Isaacs *et al*, 2013). Sporadically, the app then asks them to reflect on previous entries. An assumption is that this kind of technology-mediated reflection can increase well-being and happiness. Each reflection is shown as a stacked card with the time and a smiley happiness rating. People who used Echo reported on the many positive effects of doing so, including reliving positive experiences and overcoming negative experiences, by writing them down. The double act of recording and reflecting enabled them to generalize from the positive experiences and draw positive lessons from negative ones.

Activity 5.6

The photo on the left in [Figure 5.11](#) is of a banner placed in downtown LA, in an attempt to encourage the general public to take the stairs instead of the elevator, asking people to climb stairs on a certain day. The two photos on the right are of ambient displays (see also [Chapter 2](#)) designed to do the same thing but using more subtle and interactive methods: (i) lights that twinkled when people approach them, intended to lure them to take the stairs and (ii) clouds of different colored spheres that move up and down depending on how many people have taken the stairs or the elevator for a given period of time (gray represents elevator). The higher the orange cloud is relative to the gray one, the more people are taking the stairs than the elevator (Rogers, Hazlewood, Marshall *et al*, 2010). Which representation do you think is the most effective?

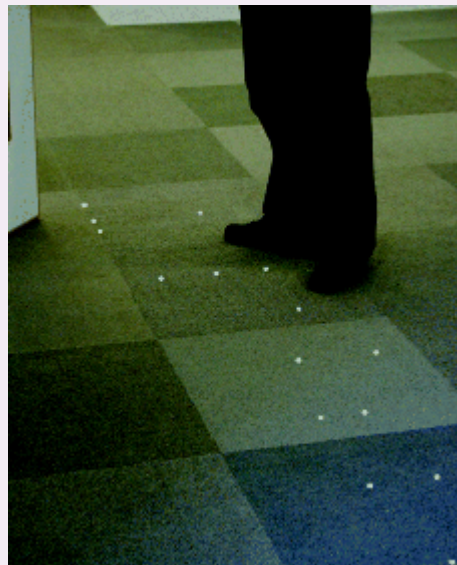


Figure 5.11 Stairs versus elevators: static sign versus ambient persuasive displays. Which is most persuasive?

Comment

Show/Hide

The global concern about climate change has led a number of HCI researchers to design and evaluate various energy sensing devices that display real-time feedback. A goal is to find ways of helping people reduce their energy consumption (and is part of a larger research agenda called sustainable HCI: see Mankoff *et al*, 2008; DiSalvo *et al*, 2010; Hazas *et al*, 2012). A focus is on persuading people to change their everyday habits with respect to environmental concerns, such as reducing their own carbon footprint, their community's (e.g. school, workplace), or an even larger organization's (e.g. street, town, country). Two early products were the Power Aware Cord and the Waatson (see [Figure 5.12](#)).



(a)



(b)

Figure 5.12 (a) The Power Aware Cord consists of an electrical power strip in which the cord is designed to visualize the energy rather than hiding it. Increase and decrease in use is conveyed through showing glowing pulses, flow, and intensity of light. (b) The Waatson (now a commercial product available in many countries) measures in watts or cost how much electricity someone is using in their home at any moment. This is conveyed in LEDs on the top side. On the underside are colored lights: when they glow blue it shows you are using less than normal; when it changes to purple it indicates that your usage is average; and when it is red it indicates you are using more than normal

Source: (a) Photo taken from the Interactive Institute's research program "Static!" and reproduced with permission. (b) Reproduced with permission from DIY Kyoto Ltd. www.diykyoto.com.

Extensive research has shown that domestic energy usage can be reduced by providing households with feedback on their consumption (Froehlich *et al*, 2010). But what are the properties that make real-time feedback effective? One dimension is frequency of feedback; continuous or daily feedback of energy consumption has been found to give higher saving results than monthly feedback. Another is the type of representation used. If it is too obvious and explicit, it may be perceived as too personal, blunt, or in your face, resulting in people objecting to it. In contrast, simple representations that are more anonymous but striking and whose function is to lure people's attention may be more effective. They may encourage people to reflect more on their energy use and even promote public debate about what is represented and how it affects them. However, if a representation is too abstract and implicit, it may be attributed other meanings, such as simply being an art piece, resulting in people ignoring it. The ideal may be somewhere in between. Peer pressure can also be effective, where peers, parents, or children chide or encourage one another to turn lights off, take a shower instead of a bath, and so on.

Another influencing factor is social norms. In a study by Schultz *et al* (2007), households were shown how their energy consumption

compared with their neighborhood average. Households above the average tended to decrease their consumption but those using less electricity than average tended to increase their consumption. The study found that this 'boomerang' effect could be counteracted by providing households with an emoticon along with the numerical information about their energy usage: households using less energy than average continued to do so if they received a smiley icon; households using more than average decreased their consumption more if they were given a sad icon.

In contrast to the Schultz study, where each household's energy consumption was kept private, the Tidy Street project (Bird and Rogers, 2010) that was run in Brighton in the UK created a large-scale visualization of the street's electricity usage by spraying a stenciled display on the road surface using chalk (see [Figure 5.13](#)). The public display was updated each day to represent how the average electricity usage of the street compared to the city of Brighton's average. The aim was to provide real-time feedback that all the householders and the general public could see change each day over a period of three weeks. The street graph also proved to be very effective at getting people who lived in Tidy Street to talk to each other about their electricity consumption and habits. It also encouraged them to talk with the many passers-by who walked up and down the street. The outcome was to reduce electricity consumption in the street by 15%, which was considerably more than other eco projects have been able to achieve.



Figure 5.13 Looking down at the Tidy Street public electricity graph from a bedroom window

BOX 5.4

The Darker Side: Deceptive Technology

Technology is increasingly being used to deceive people into parting with their personal details that allow Internet fraudsters to access their bank accounts and draw money from them.

Authentic-looking letters, appearing to be sent from eBay, PayPal, and various leading banks, are spammed across the world, ending up in people's email boxes with messages such as 'During our regular verification of accounts, we couldn't verify your information. Please click here to update and verify your information.' Given that many people have an account with one of these corporations, there is a chance that they will be misled and unwittingly follow what is being asked of them, only to discover a few days later they are several thousand dollars worse off.

Similarly, letters from supposedly super-rich individuals in far-away countries, offering a share of their assets if the recipient of the email provides them with his bank details, have been persistently spammed worldwide. While many people are becoming increasingly wary of what are known as phishing scams, there are still many vulnerable people who are gullible to such tactics. (Note: The term phishing is a play on the term fishing that refers to the sophisticated way of luring users' financial information and passwords.) Moreover, Internet fraudsters are becoming smarter and are always changing their tactics. While the art of deception is centuries old, the increasing, pervasive, and often ingenious use of the web to trick people into

divulging personal information may have catastrophic effects on society. ■

5.7 Anthropomorphism and Zoomorphism

Anthropomorphism is the propensity people have to attribute human qualities to animals and objects while zoomorphism is the shaping of an object or design in animal form. For example, people sometimes talk to their computers as if they were humans, treat their robot cleaners as if they were their pets, and give all manner of cute names to their mobile devices, routers, and so on. Advertisers are well aware of these phenomena and often create human-like and animal-like characters out of inanimate objects to promote their products. For example, breakfast cereals, butter, and fruit drinks have all been transmogrified into characters with human qualities (they move, talk, have personalities, and show emotions), enticing the viewer to buy them. Children are especially susceptible to this kind of magic, as witnessed by their love of cartoons, where all manner of inanimate objects are brought to life with human-like qualities.

The finding that people, especially children, have a propensity to accept and enjoy objects that have been given human-like qualities has led many designers to capitalize on it, most notably in the design of human–computer dialogs modeled on how humans talk to each other. It is now possible to have conversations and interact with various screen characters, such as agents, tutors, and virtual pets, as if they were human.

Anthropomorphism has also been exploited in the development of computer-based cuddly toys that are embedded with various sensors. Early commercial products like ActiMates™ were designed to encourage children to learn through playing with them. Barney (a bear), for example, attempted to motivate play in children by using human-based speech and movement (Strommen, 1998). The toys were programmed to react to the child and make comments while watching TV together or working together on a computer-based task.

In particular, Barney was programmed to congratulate the child whenever she produced a right answer and also to react to the content on screen with appropriate emotions, e.g. cheering at good news and expressing concern at bad news. Interactive dolls have been designed to talk, sense, and understand the world around them, using sensor-based technologies, speech recognition, and various mechanical protractors embedded in their bodies. For example, Amazing Amanda was able to exhibit a number of facial expressions to convey her feelings. If she was offered something to eat she did not want, e.g. a piece of plastic pizza embedded with an RFID tag that when placed near her mouth was read by a tag reader hidden in her neck, she would contort her face and say 'I don't want that.'

Video of Amazing Amanda in Action at
<http://youtu.be/1gJA3NkmlUY>

Furnishing technologies with personalities and other human-like attributes makes them more enjoyable and fun to interact with. They can also motivate people to carry out various activities, such as learning. Being addressed in the first person (e.g. 'Hello Chris! Nice to see you again. Welcome back. Now what were we doing last time? Oh yes, exercise 5. Let's start again') is much more appealing than being addressed in the impersonal third person ('User 24, commence exercise 5'), especially for children. It can make them feel more at ease and reduce their anxiety. Similarly, interacting with screen characters like tutors and wizards can be much more pleasant than interacting with a cold dialog box or blinking cursor on a blank screen.

However, virtual agents, robots, and toys can also be patronizing and annoying in certain contexts. They also do not have the range of emotional intelligence to respond in the nuanced ways humans do with each other. Nevertheless much effort has gone into designing interface agents to be life-like, exhibiting realistic human movements, like walking and running, and having distinct personalities and traits. The design of the characters' appearance, their facial expressions,

and how their lips move when talking are all considered important interface design concerns. This has included modeling various conversational mechanisms such as:

- Recognizing and responding to verbal and non-verbal input.
- Generating verbal and non-verbal output.
- Coping with breakdowns, turn-taking, and other conversational mechanisms.
- Giving signals that indicate the state of the conversation as well as contributing new suggestions for the dialog (Cassell, 2000).

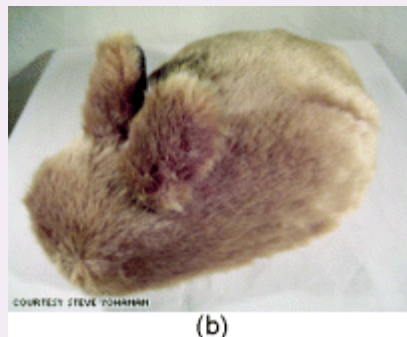
Activity 5.7

A Robot or A Cuddly Pet?

Early robot pets, such as Sony's AIBO and QRIO, were made of metal and looked like robots. In contrast, a more recent trend has been to make robot pets feel and look more like real pets by covering them in fur (e.g. squirrels, cats, rabbits) and making them behave in more cute pet-like ways. Two contrasting examples are presented in [Figure 5.14](#). Which do you prefer and why?



(a)



(b)

Figure 5.14 Two kinds of robot pets

Source: (a) Courtesy of Sony Corporation, (b) Reproduced with permission of Steve Yohanan. **Photo:** Martin Dee.

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Audio of a radio show recorded in 2014 about emotions and emotional technology with interaction designers Jennifer Dunnam, Don Norman, and Elizabeth Churchill at <http://www.cbc.ca/spark/blog/2014/06/08/emotional-tech/>

Assignment

This assignment requires you to write a critique of the persuasive impact of a virtual agent by considering what it would take for a virtual agent to be believable, trustworthy, and convincing.

- a. Look at a website that has a virtual assistant, e.g. Anna at [ikea.com](https://www.ikea.com), Jenn at [Alaskaair.com](https://www.alaskaair.com) and answer the following:
- What does the virtual agent do?
 - What type of agent is it?
 - Does it elicit an emotional response from you? If so, what kind?
 - What kind of personality does it have?
 - How is this expressed?
 - What kinds of behavior does it exhibit?
 - What are its facial expressions like?
 - What is its appearance like? Is it realistic or cartoon-like?
 - Where does it appear on the screen?
 - How does it communicate with the user (text or speech)?
 - Is the level of discourse patronizing or at the right level?
 - Is the agent helpful in guiding the user towards making a purchase or finding out something?
 - Is it too pushy?
 - What gender is it? Do you think this makes a difference?
 - Would you trust the agent to the extent that you would be happy to buy a product from it or follow its guidance? If not, why not?
 - What else would it take to make the agent persuasive?

b. Next, look at an equivalent website that does not include an agent but is based on a conceptual model of browsing, e.g. [Amazon.com](https://www.amazon.com). How does it compare with the agent-based site you have just looked at?

- Is it easy to find information?
- What kind of mechanism does the site use to make recommendations and guide the user in making a purchase or finding out information?
- Is any kind of personalization used at the interface to make the user feel welcome or special?
- Would the site be improved by having an agent? Explain your reasons either way.

c. Finally, discuss which site you would trust most and give your reasons for this.

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Summary

This chapter has described the different ways interactive products can be designed (both deliberately and inadvertently) to make people respond in certain ways. The extent to which users will learn, buy a product online, quit a bad habit, or chat with others depends on how convincing the interface is, how comfortable they feel when using a product, or how much they can trust it. If the interactive product is frustrating to use, annoying, or patronizing, users will easily become angry and despondent, and often stop using it. If, on the other hand, the product is pleasurable, enjoyable to use, and makes people feel comfortable and at ease, then they will continue to use it, make a purchase, return to the website, or continue to learn. This chapter has described various interaction mechanisms that can be used to elicit positive emotional responses in users and ways of avoiding negative ones.

Key points

- Emotional aspects of interaction design are concerned with how to facilitate certain states (e.g. pleasure) or avoid certain reactions (e.g. frustration) in user experiences.
- Well-designed interfaces can elicit good feelings in people.
- Aesthetically pleasing interfaces can be a pleasure to use.
- Expressive interfaces can provide reassuring feedback to users as well as be informative and fun.
- Badly designed interfaces often make people frustrated, annoyed, or angry.
- Emotional technologies can be designed to persuade people to change their behaviors or attitudes.
- Anthropomorphism is the attribution of human qualities to objects.

- Virtual agents and robot pets have been developed to make people feel motivated, reassured, and in a good mood.

Further Reading

Fogg, B. J. (2003) *Persuasive Technology: Using computers to change what we think and do*. Morgan Kaufmann. This is a very readable and provocative book, explaining how a diversity of technologies can and have been designed to persuade people to change their behavior and attitudes. It presents a conceptual framework of the different types and a host of examples, together with discussing social, ethical, and credibility issues to do with using persuasive technologies.

Jordan, P. W. (2000) *Designing Pleasurable Products*. Taylor & Francis. This book was written primarily for a product design audience to consider as part of the human factors. However, its applicability to interaction design has meant that it has become a popular book for those wanting to understand more about the relationship between usability and pleasure. It provides many illuminating case studies of the design of products, such as cars, cameras, and clocks. It also provides detailed product benefits specifications that are a form of guidance on how to design and evaluate pleasurable aspects.

LEDOUX, J. E. (1998) *The Emotional Brain: The mysterious underpinnings of emotional life*. Simon & Schuster. This book explains what causes us to feel fear, love, hate, anger, and joy and explores whether we control our emotions versus them controlling us. The book also covers the origins of human emotions and explains that many evolved to enable us to survive.

Norman, D. (2005) *Emotional Design: Why we love (or hate) everyday things*. Basic Books. This book is an easy read while at the same time being thought-provoking. We get to see inside Dan Norman's kitchen and learn about the design aesthetics of his collection of teapots. The book also includes essays on the

emotional aspects of robots, computer games, and a host of other pleasurable interfaces.

WALTER, A. (2011) *A Book Apart: Designing for Emotion*. Zeldman, Jeffrey. This short book is targeted at web designers who want to understand how to design websites that users will enjoy and want to come back to. It covers the classic literature on emotions and proposes practical approaches to emotional web design.